

SAGNIK DATTA

Full Stack Developer | AI/ML Engineer | Software Engineer

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PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate (B.Tech, VIT Vellore) specializing in **full-stack development**, **AI/ML engineering**, and **cloud-ready backend systems**. Proficient in **Python**, **JavaScript/TypeScript**, and **C++** with hands-on experience building **production-grade REST APIs**, integrating **large language models (LLMs)**, and deploying **scalable microservices architectures**. Demonstrated ability to deliver end-to-end software solutions — from database schema design to responsive front-end interfaces. Actively seeking **software engineering** or **AI/ML internship** roles to drive impactful, data-driven product development.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- **Backend:** FastAPI, Node.js, Express.js, RESTful API Design, GraphQL, JWT, OAuth 2.0, Microservices, Web-Sockets
- **Frontend:** React.js, Next.js, Redux, Tailwind CSS, Bootstrap, Responsive & Mobile-First Design
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLAlchemy (Async), Database Design, Query Optimization
- **AI/ML & NLP:** LLM Integration (OpenAI, Groq API, LLaMA 3), Prompt Engineering, OCR, TensorFlow, Scikit-learn, NLP Pipelines, RAG, Data Preprocessing
- **DevOps & Tools:** Git, GitHub, Docker, CI/CD Pipelines, Alembic, ReportLab, Agile/Scrum, UML Diagrams, Swagger/OpenAPI
- **Cloud & Deployment:** AWS, Containerization (Docker), Cloud Deployment, Virtual Environments

WORK EXPERIENCE

Front-End Software Engineering Intern

Skyscanner via Forage — Remote

February 2026 – March 2026

- Engineered a fully responsive **Backpack React** web application using **component-based architecture** and modern UI/UX principles, reducing UI inconsistencies by standardizing reusable functional components.
- Implemented **React Hooks** and **state management** patterns (useState, useEffect) to power real-time interactive features, improving component reusability across the codebase.
- Applied **mobile-first responsive design** ensuring cross-browser compatibility; improved page performance through **code splitting** and **lazy loading**, reducing initial load time.
- Collaborated in an **Agile/Scrum** environment, participating in sprint planning, code reviews, and iterative delivery cycles.

Software Engineer Intern

Electronic Arts (EA) via Forage — Remote

April 2025 – May 2025

- Designed and prototyped the **“Rewind Dynasty Mode”** feature for an EA Sports title, enhancing gameplay flexibility and projected to boost user retention metrics.
- Created detailed **UML class diagrams** and implemented core game logic in **C++** using **object-oriented design** principles (inheritance, polymorphism, encapsulation).
- Identified and resolved **5+ critical software bugs** through systematic debugging, improving system stability and reducing crash frequency.
- Applied **SDLC methodologies** and **version control** (Git) best practices within a cross-functional Agile team environment.

PROJECTS

AI Resume Builder API *FastAPI · PostgreSQL · Groq LLM (LLaMA 3.3 70B) · ReportLab · JWT · SQLAlchemy*
github.com/sagnikdatta2k6/airesumecontentbuilderAPI

- Architected and deployed a **production-ready REST API** with **JWT authentication**, **async SQLAlchemy ORM**, and **PostgreSQL** — handling secure CRUD operations with Alembic database migrations.
- Integrated **Groq’s LLaMA 3.3 70B LLM** for AI-powered content generation (professional summaries, tailored bullet points, cover letters) using advanced **prompt engineering** and NLP techniques.
- Built an intelligent **job description matching engine** that tailors resumes to target JDs, improving **ATS compatibility** and keyword alignment scores.
- Implemented **PDF export** with professional formatting via ReportLab; deployed with **rate limiting**, **CORS**, and **comprehensive Swagger/OpenAPI** documentation for developer adoption.
- Enforced security best practices: **input validation**, **error handling**, **OAuth flows**, and environment-based configuration management.

Real-Time Crypto Arbitrage Detection System

Python · WebSockets · REST APIs · Async Programming

- Built a **real-time cryptocurrency monitoring system** tracking live prices across **multiple exchanges simultaneously** using concurrent WebSocket connections for **low-latency data streaming**.
- Engineered a **multi-exchange spread detection algorithm** achieving **sub-second latency** for time-sensitive arbitrage opportunity identification and automated alerting.
- Optimized **asynchronous data processing pipelines** to handle high-frequency tick data with minimal memory overhead and maximum throughput.

Noto.ai — AI-Powered Study Toolkit *Python · OCR · NLP · LLM Integration · RAG*

- Engineered an **AI-powered study platform** with **OCR capabilities** for document digitization, enabling automated intelligent text extraction from handwritten and printed materials.
- Implemented **LLM-driven context-aware summarization** and **Retrieval-Augmented Generation (RAG)** for knowledge organization and study material generation.
- Applied **NLP pipelines** for automated note-taking, topic clustering, and key concept extraction to enhance student productivity.

Rock-Paper-Scissors Adaptive AI *Python · LLaMA 3 · Streamlit · Machine Learning*

- Developed an **adaptive AI game agent** using **real-time machine learning** and **pattern recognition** to predict opponent strategies and dynamically improve win rate across sessions.
- Deployed an **interactive Streamlit web interface** with real-time game state management and visual feedback, demonstrating full-stack ML deployment skills.

than0s — Discord AI Assistant Bot *Discord.py · NLP · API Integration*

- Built an **AI-powered Discord bot** using **NLP and multi-API integration**, delivering intelligent contextual responses, automated command handling, and stateful conversation management for community servers.

File Utilities Suite *Python · File Processing · Algorithm Design*

- Developed **anyFileToPDF**: a multi-format file conversion utility handling diverse file types to PDF with consistent output fidelity.
- Created **fileCompressor**: an efficient lossless compression tool reducing file sizes significantly using optimized compression algorithms.

EDUCATION

B.Tech in Computer Science & Engineering (IoT Specialization) *2024 – Present*

Vellore Institute of Technology (VIT), Vellore, Tamil Nadu
CGPA: In Progress | Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, ML Fundamentals

Class XII — Computer Science (CBSE) *2024*

G D Goenka Public School, Dakshineswar, West Bengal

Class X — Science (ICSE) *2022*

Don Bosco School, Liluah, West Bengal

CERTIFICATIONS

- **Frontend Software Engineering Job Simulation** — Skyscanner via Forage, 2026
- **Software Engineering Job Simulation** — Electronic Arts (EA) via Forage, 2025
- **Python: Mastering The Essentials** — Scaler Academy, 2025
- **Python: Data Structures & Algorithms** — Great Learning, 2025
- **Generative AI for Beginners** — Great Learning, 2025

ACHIEVEMENTS & HONORS

- **Reverse Coding Competition 2026 (ACM-VIT)**: Ranked in **Top 36 of 200+ teams** — Top 18% at VIT Vellore.
- **Reverse Coding Competition 2025 (ACM-VIT)**: Ranked in **Top 50 of 180 teams** — Top 28% at VIT Vellore.
- **Hackathons**: Regular participant in university-level hackathons and competitive coding contests; consistent top-percentile finishes.
- **Open Source Contributor**: Actively maintains and contributes to open-source projects on GitHub, demonstrating collaborative development practices.

LANGUAGES

English (Professional Proficiency) | Bengali (Native) | Hindi (Professional Proficiency)